

ALFAAJ61459

## Yeast lysis solution for DNA isolation

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:** 酵母裂解液 溶液 用于 DNA 分离  
**Product Description:** Yeast lysis solution for DNA isolation

**Cat No. :** J61459

**Supplier** Avocado Research Chemicals Ltd.  
(Part of Thermo Fisher Scientific)  
Shore Road, Heysham  
Lancashire, LA3 2XY,  
United Kingdom  
Office Tel: +44 (0) 1524 850506  
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**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

|                                                                                          |                                |                                         |
|------------------------------------------------------------------------------------------|--------------------------------|-----------------------------------------|
| <b>Physical State</b><br>Liquid Viscous liquid                                           | <b>Appearance</b><br>Colorless | <b>Odor</b><br>No information available |
| <b>Emergency Overview</b><br>Causes mild skin irritation. Causes serious eye irritation. |                                |                                         |

#### Classification of the substance or mixture

|                                   |            |
|-----------------------------------|------------|
| Skin Corrosion/Irritation         | Category 3 |
| Serious Eye Damage/Eye Irritation | Category 2 |

#### Label Elements



**Signal Word**

**Warning**

#### **Hazard Statements**

H316 - Causes mild skin irritation  
H319 - Causes serious eye irritation

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### Precautionary Statements

#### Prevention

P264 - Wash face, hands and any exposed skin thoroughly after handling

P280 - Wear eye protection/ face protection

#### Response

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P337 + P313 - If eye irritation persists: Get medical advice/attention

#### Storage

P403 - Store in a well-ventilated place

#### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

### Physical and Chemical Hazards

None identified.

### Health Hazards

Causes mild skin irritation. Causes serious eye irritation.

### Environmental hazards

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

Contains a known or suspected endocrine disruptor. Included in the list established in accordance with Article 59(1) for having endocrine disrupting properties. Contains a substance on the National Authorities Endocrine Disruptor Lists.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

| Component                                                                               | CAS No    | Weight % |
|-----------------------------------------------------------------------------------------|-----------|----------|
| Water                                                                                   | 7732-18-5 | 93.76    |
| Ethylenediaminetetraacetic acid, disodium salt dihydrate                                | 6381-92-6 | 2.33     |
| Lithium chloride                                                                        | 7447-41-8 | 2.12     |
| Poly(oxy-1,2-ethanediyl), .alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-.omega.-hydroxy- | 9002-93-1 | 1        |
| 1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, hydrochloride                              | 1185-53-1 | 0.79     |

## SECTION 4. FIRST AID MEASURES

### General Advice

If symptoms persist, call a physician.

### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

### Ingestion

Clean mouth with water and drink afterwards plenty of water.

### Most important symptoms and effects

None reasonably foreseeable.

### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### Notes to Physician

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Carbon dioxide (CO<sub>2</sub>). Powder. Water spray. In case of major fire and large quantities: Evacuate area. Fight fire remotely due to the risk of explosion.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required.

**Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information. Do not allow material to contaminate ground water system. Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

**Storage**

Keep refrigerated.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Monitoring methods**

MDHS 91 Metals and metalloids in workplace air by X-ray fluorescence spectrometry MDHS 99 Metals in air by ICP-AES

**Exposure Controls****Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release

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or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

### Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                   |                          |                                          |
|-----------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>                 | Colorless                |                                          |
| <b>Physical State</b>             | Liquid Viscous liquid    |                                          |
| <b>Odor</b>                       | No information available |                                          |
| <b>Odor Threshold</b>             | No data available        |                                          |
| <b>pH</b>                         | No information available |                                          |
| <b>Melting Point/Range</b>        | No data available        |                                          |
| <b>Softening Point</b>            | No data available        |                                          |
| <b>Boiling Point/Range</b>        | No information available |                                          |
| <b>Flash Point</b>                | No information available | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>           | No data available        |                                          |
| <b>Flammability (solid,gas)</b>   | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>           | No data available        |                                          |
| <b>Vapor Pressure</b>             | No data available        |                                          |
| <b>Vapor Density</b>              | No data available        | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b> | No data available        |                                          |
| <b>Bulk Density</b>               | Not applicable           | Liquid                                   |
| <b>Water Solubility</b>           | Miscible                 |                                          |

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|                                                                                                                |                          |
|----------------------------------------------------------------------------------------------------------------|--------------------------|
| <b>Solubility in other solvents</b>                                                                            | No information available |
| <b>Partition Coefficient (n-octanol/water)</b>                                                                 |                          |
| <b>Component</b>                                                                                               | <b>log Pow</b>           |
| Lithium chloride                                                                                               | -2.66                    |
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-.omega.-hydroxy-<br>1,3-Propanediol, | 2.7                      |
| 2-amino-2-(hydroxymethyl)-,<br>hydrochloride                                                                   | -3.6                     |
| <b>Autoignition Temperature</b>                                                                                | No data available        |
| <b>Decomposition Temperature</b>                                                                               | No data available        |
| <b>Viscosity</b>                                                                                               | No data available        |
| <b>Explosive Properties</b>                                                                                    | No information available |
| <b>Oxidizing Properties</b>                                                                                    | No information available |

### SECTION 10. STABILITY AND REACTIVITY

|                                 |                                 |
|---------------------------------|---------------------------------|
| <b>Stability</b>                | Stable under normal conditions. |
| <b>Hazardous Reactions</b>      | None under normal processing.   |
| <b>Hazardous Polymerization</b> | No information available.       |
| <b>Conditions to Avoid</b>      | None known.                     |
| <b>Materials to avoid</b>       | No information available.       |

**Hazardous Decomposition Products** Nitrogen oxides (NOx). Hydrogen chloride. Lithium oxide. Sodium oxides.

### SECTION 11. TOXICOLOGICAL INFORMATION

#### Product Information

##### (a) acute toxicity; Toxicology data for the components

| Component                                                                                                          | LD50 Oral                              | LD50 Dermal                            | LC50 Inhalation     |
|--------------------------------------------------------------------------------------------------------------------|----------------------------------------|----------------------------------------|---------------------|
| Water                                                                                                              | -                                      | -                                      | -                   |
| Lithium chloride                                                                                                   | LD50 = 526 mg/kg ( Rat )               | >2000 mg/kg (Rat)                      | >5.57 mg/L/4h (Rat) |
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-<br>.omega.-hydroxy-<br>1,3-Propanediol, | 1800 mg/kg ( Rat )                     |                                        |                     |
| 2-amino-2-(hydroxymethyl)-, hydrochloride                                                                          | OECD 425 (Rat)<br>LD50 > 5000 mg/kg bw | OECD 402 (Rat)<br>LD50 > 5000 mg/kg bw |                     |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; Category 2

##### (d) respiratory or skin sensitization;

|                    |                   |
|--------------------|-------------------|
| <b>Respiratory</b> | No data available |
| <b>Skin</b>        | No data available |

| Component                                                     | Test method             | Test species | Study result    |
|---------------------------------------------------------------|-------------------------|--------------|-----------------|
| 1,3-Propanediol,<br>2-amino-2-(hydroxymethyl)-, hydrochloride | OECD Test Guideline 406 | guinea pig   | non-sensitising |

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|                    |  |  |  |
|--------------------|--|--|--|
| 1185-53-1 ( 0.79 ) |  |  |  |
|--------------------|--|--|--|

(e) germ cell mutagenicity; No data available

| Component                                                                           | Test method                                                | Test species          | Study result |
|-------------------------------------------------------------------------------------|------------------------------------------------------------|-----------------------|--------------|
| 1,3-Propanediol,<br>2-amino-2-(hydroxymethyl)-, hydrochloride<br>1185-53-1 ( 0.79 ) | OECD Test Guideline 471<br>Bacterial Reverse Mutation Test | Mammalian<br>in vitro | negative     |

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed No information available

## SECTION 12. ECOLOGICAL INFORMATION

**Ecotoxicity effects** May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component                                                                                      | Freshwater Fish                                                     | Water Flea                            | Freshwater Algae | Microtox                          |
|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------|------------------|-----------------------------------|
| Lithium chloride                                                                               | EC50: 158 mg/L/96h<br>(rainbow trout)                               |                                       |                  |                                   |
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-<br>.omega.-hydroxy- | LC50 = 8.9 mg/L 96H<br>LC50 = 4.0 mg/l 96H<br>(Pimephales promelus) | EC50 = 26 mg/L 48h                    | -                | -                                 |
| 1,3-Propanediol,<br>2-amino-2-(hydroxymethyl)-, hydrochloride                                  |                                                                     | Daphnia Magna<br>EC50 >100 mg/L (48h) |                  | OECD 209<br>EC50 > 1000 mg/L (3h) |

**Persistence and Degradability** Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary based on information available, May persist.

### Persistence

| Component                                                                                                     | Degradability |
|---------------------------------------------------------------------------------------------------------------|---------------|
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-.omega.-hydroxy-<br>9002-93-1 ( 1 ) | 60% >28 days  |

### Degradation in sewage treatment plant

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential** May have some potential to bioaccumulate

| Component                                                                                      | log Pow | Bioconcentration factor (BCF) |
|------------------------------------------------------------------------------------------------|---------|-------------------------------|
| Lithium chloride                                                                               | -2.66   | No data available             |
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-<br>.omega.-hydroxy- | 2.7     | No data available             |

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|                                                               |      |                   |
|---------------------------------------------------------------|------|-------------------|
| 1,3-Propanediol,<br>2-amino-2-(hydroxymethyl)-, hydrochloride | -3.6 | No data available |
|---------------------------------------------------------------|------|-------------------|

**Mobility in soil**

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils

**Endocrine Disruptor Information**

| Component                                                                                      | EU - Endocrine Disruptors<br>Candidate List | EU - Endocrine Disruptors -<br>Evaluated Substances | Japan - Endocrine Disruptor<br>Information |
|------------------------------------------------------------------------------------------------|---------------------------------------------|-----------------------------------------------------|--------------------------------------------|
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-<br>.omega.-hydroxy- | Group III Chemical                          | -                                                   | -                                          |

**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
 This product does not contain any known or suspected substance

### SECTION 13. DISPOSAL CONSIDERATIONS

**Waste from Residues/Unused  
Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**Other Information**

Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

### SECTION 14. TRANSPORT INFORMATION

**Road and Rail Transport**

Not Regulated

**IMDG/IMO**

Not regulated

**IATA**

Not regulated

**Special Precautions for User**

No special precautions required

### SECTION 15. REGULATORY INFORMATION

**International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component                                                                                  | The<br>Inventory of<br>Hazardous<br>Chemicals<br>(2015<br>Edition) | List of<br>dangerous<br>goods GB<br>12268 -<br>2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Water                                                                                      | -                                                                  | -                                                   | X    | X     | 231-791-2 | X    | X   | X     | X    |      | X    | KE-35400 |
| Ethylenediaminetetraacetic acid, disodium salt dihydrate                                   | -                                                                  | -                                                   | X    | X     | -         | -    | X   | X     | -    |      | X    | -        |
| Lithium chloride                                                                           | -                                                                  | -                                                   | X    | X     | 231-212-3 | X    | X   | X     | X    | X    | X    | KE-22552 |
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-.omega.-hydroxy- | -                                                                  | -                                                   | X    | X     | -         | X    | X   | X     | X    | X    | X    | KE-33568 |

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|                                                                                     |   |   |   |   |           |   |   |   |   |  |   |          |
|-------------------------------------------------------------------------------------|---|---|---|---|-----------|---|---|---|---|--|---|----------|
| mega.-hydroxy-<br>1,3-Propanediol,<br>2-amino-2-(hydroxyme<br>thyl)-, hydrochloride | - | - | X | X | 214-684-5 | X | X | X | X |  | X | KE-34819 |
|-------------------------------------------------------------------------------------|---|---|---|---|-----------|---|---|---|---|--|---|----------|

### National Regulations

## SECTION 16. OTHER INFORMATION

**Prepared By** Health, Safety and Environmental Department  
**Revision Date** 09-May-2024  
**Revision Summary** New emergency telephone response service provider.

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Physical hazards**

On basis of test data

**Health Hazards**

Calculation method

**Environmental hazards**

Calculation method

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage,

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transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**